Code Functionality and Logic

The script executes a sequential data preparation pipeline, which can be broken down into five key stages:

1. Data Loading and Initial Assessment

The process begins by loading the `train.csv` dataset into a pandas DataFrame. An initial assessment is performed using the `.info()` and `.isnull().sum()` methods to understand the data types of each column and to identify the extent of missing values across the dataset. This step is crucial for diagnosing data quality issues that need to be addressed.

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2. Handling Missing Values (Imputation)

To create a complete dataset, missing values were handled using appropriate imputation strategies based on the data type of the feature:

- **Categorical Features:** For columns such as `Gender`, `Married`, `Dependents`, `Self_Employed`, and `Credit_History`, missing values were filled using the **mode** (the most frequently occurring value in each column). This is a standard practice for categorical data as it replaces missing entries with the most probable category.
- **Numerical Features:** For numerical columns like `LoanAmount` and `Loan_Amount_Term`, missing values were imputed using the **median** of each column. The median was chosen over the mean because it is more robust to outliers and skewed data, providing a more representative measure of central tendency for these features.

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3. Data Correction and Type Conversion

A critical data cleaning step involved correcting the `Dependents` column. This feature contained a non-numeric value, `'3+'`, which would cause errors during model training. This value was programmatically replaced with `3`, and the entire column was then converted to a numeric data type, ensuring all its values are integers.

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4. Feature Engineering

To enhance the predictive power of the model, two feature engineering techniques were applied:

- **Creation of `Total_Income`:** A new feature, `Total_Income`, was created by summing the `ApplicantIncome` and `CoapplicantIncome` columns. This provides a more holistic measure of a household's financial capacity to repay a loan.
- **Logarithmic Transformation:** To normalize the distribution of skewed numerical features (`ApplicantIncome`, `LoanAmount`, `Total_Income`, etc.), a **logarithmic transformation** (`np.log1p`) was applied. This technique helps to mitigate the effect of extreme values (outliers) and can significantly improve the performance of linear models like logistic regression.

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5. Encoding Categorical Variables

Finally, the script converts all categorical features into a numerical format that can be processed by machine learning algorithms:

- **Binary Mapping:** Categorical features with only two distinct values (e.g., `Gender`, `Married`, `Loan_Status`) were mapped to binary integers (`1` and `0`).
- **One-Hot Encoding:** For the `Property_Area` feature, which contains multiple nominal categories ("Urban", "Rural", "Semiurban"), **one-hot encoding** was used. This method creates new binary columns for each category, preventing the model from incorrectly assuming an ordinal relationship between them.

The final output of this script is a clean, fully numeric DataFrame, ready for the subsequent stages of model training and evaluation.